

Xi Huang

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Short Bio

Robotist specializing in enabling robots to interact with dynamic, cluttered environments. Skilled in developing **machine learning-driven algorithms** for robot **motion planning and manipulation** informed by sensor perception, **computer vision** and **3D scene understanding**. Proven ability to deploy state-of-the-art methods to real-world robotic systems. Active contributor to open-source robotics projects.



Research interests:

Motion planning, Robot Manipulation, Scene Understanding, Multi-Robot Systems, Optimization, Embodiment Transfer

Skills

Programming: C++, Python, ROS, PyBullet, Mujoco, IsaacSim, Gazebo, MATLAB

Machine Learning

- **Frameworks:** PyTorch, TensorFlow, PyTorch-Lightning, scikit-learn, WandB, Hydra, Gym, Slurm
- **Expertise:** Reinforcement Learning, Imitation Learning, 3D Scene Understanding (NeRFs, Gaussian Splatting), Sequential Models, Vision-Language Models

Optimization & Control: Bayesian Methods, Convex Optimization, Model Predictive Control (MPC), Force-based Manipulation, Learning-based Control

Education

Karlsruhe Institute of Technology, Ph.D. Candidate in Computer Science 02/2019 – Present

- Designed algorithms using **imitation learning**, **reinforcement learning** and **NeRFs** for robot motion planning and manipulation, validated on diverse real-world embodiments
- Built a comprehensive **framework for machine learning research on robotics**, integrating simulation, data management, model training, and real-world deployment.
- Contributed publications to top-tier peer-reviewed robotics conferences (IROS, ICRA) and journals (RAL, RAS)
- **Led three industrial/research projects** focusing on real-time motion generation and robot safety
- **Dissertation:** Online Motion Planning for Robot Manipulators in Dynamic Environments (Defense anticipated: late 2025, advised by [Torsten Kröger](#) and [Tamim Asfour](#))

Karlsruhe Institute of Technology, M.Sc. in Mechanical Engineering 10/2015 – 11/2018

- Specialized in **robotics** and **control theory**
- Thesis: Force-Based Process Control for Automated Grinding with an Industrial Robot

South China University of Technology, B.Sc. in Vehicle Engineering 09/2010 – 06/2014

- Specialized in **electronic vehicles** and **driving dynamics**
- Thesis: Modeling and Simulation of Driving Cycles for Fuel-Cell Vehicle Optimization

Experience

Robotics Researcher, Stogl Robotics, Part-time – Karlsruhe, Germany 01/2023 – Present

- Developed robot motion planning and manipulation pipeline for a mobile manipulator, incl. algorithms, visualization and control
- Extended ROS2 control interfaces for efficient real-world deployment

Control Algorithm Intern, IAV – Gifhorn, Germany

08/2017 - 01/2018

- Data processing and feature extraction, finding latent representation of data regarding vehicle dynamics
- Designed, implemented control algorithms for the latent model grounded on data, and validated on real systems

Research Assistant, KIT – Karlsruhe, Germany

12/2015 – 10/2018

- Institute of Robotics (IAR): Applied machine learning to recognize and classify human action, enabling seamless human-robot interaction for tasks such as object handovers.
- Institute of Fluid Mechanics (ISTM): Developed numerical optimization methods for turbulence modeling, enhancing the accuracy of computational fluid dynamics simulations.

Publications

X-IL: Exploring the Design Space of Imitation Learning Policies	Under review
Xiaogang Jia, Xi Huang , Atalay Donat, Xuan Zhao, Denis Blessing, Hongyi Zhou, Han A. Wang, Qian Wang, Rudolf Lioutikov, Gerhard Neumann	preprint
Towards Fusing Point Cloud and Visual Representations for Imitation Learning	Under review
Atalay Donat, Xiaogang Jia, Xi Huang , Aleksandar Taranovic, Denis Blessing, Ge Li, Hongyi Zhou, Rudolf Lioutikov, Gerhard Neumann	preprint
MoRe-ERL: Learning Motion Residuals using Episodic Reinforcement Learning	Under review
Xi Huang , Hongyi Zhou, Ge Li, Yucheng Tang, Björn Hein, Tamim Asfour and Rudolf Lioutikov	
ETA-IK: Execution-Time-Aware Inverse Kinematics for Dual-Arm Systems	Under review
Xi Huang , Yucheng Tang, Ilshat Mamaev, and Björn Hein	preprint
dGrasp: NeRF-Informed Implicit Grasp Policies with Supervised Optimization Slopes	RAS
Gergely Sóti, Xi Huang , Björn Hein	preprint
Planning with Learned Subgoals Selected by Temporal Information	ICRA 2024
Xi Huang , Gergely Sóti, Christoph Ledermann, Björn Hein and Torsten Kröger	preprint
6-DoF Grasp Pose Evaluation and Optimization via Transfer Learning from NeRFs	ICRA 2024
Gergely Sóti, Xi Huang , Björn Hein	preprint
Train What You Know: Precise Pick-and-Place with Transporter Networks	ICRA 2023
Gergely Sóti, Xi Huang , Björn Hein	preprint
HIRO: Heuristics Informed Robot Online Path Planning using Pre-computed Deterministic Roadmaps	IROS 2022
Xi Huang , Gergely Sóti, Christoph Ledermann, Björn Hein and Torsten Kröger	preprint

Open-source Projects

Real-Time Time-Optimal Trajectory Parameterization	repository
• A real-time-capable algorithm that can handle a discrete set of path waypoints and reliably generate trajectories within hard real-time constraints less than 1 ms and hard kinematic constraints regarding joint velocity and acceleration	